User manual

This is the official user manual for the Unity editor tool Spline Architect (base package). Documentation for optional addons is provided in a separate Addons Manual.

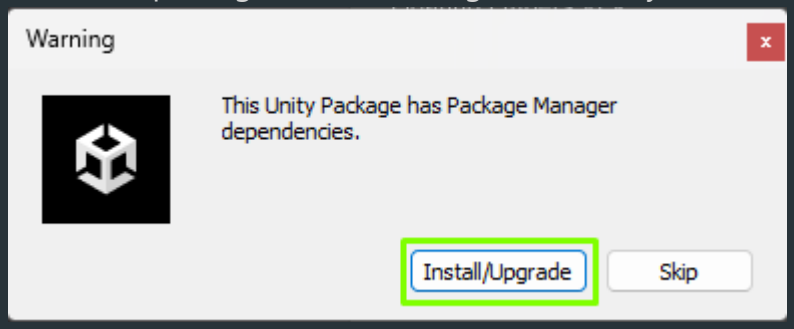
> Having trouble migrating to Spline Architect 2.1.0? See the Migration Guide

Installation

> Works with **Unity 2022** and **6000**.

> ! Minimum required version is **Unity 2022.3.62**.

Open your unity editor and go to Window → Package Manager → My Assets, and install Spline Architect. After that you need to press Install/Upgrade in the Warning window. This will install the collection package (or do nothing if its already installed).



Updating

> ! If you encounter errors after updating, remove the entire Spline Architect folder and re-import the package(s). Also check for updates to any of your addons!

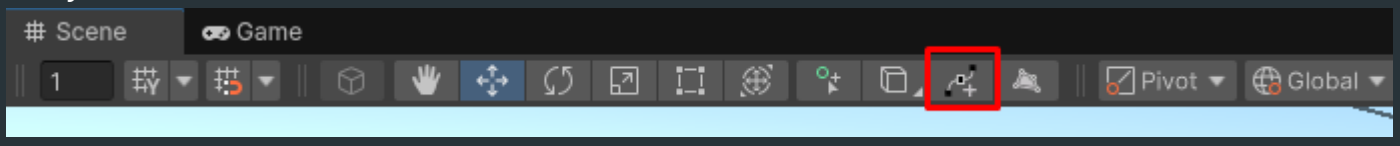
To update, open Unity and go to Window → Package Manager → My Assets, then update the related package. After updating, you'll see the same Warning window that appeared when first installing Spline Architect. Press Install/Upgrade.

Create splines and control points

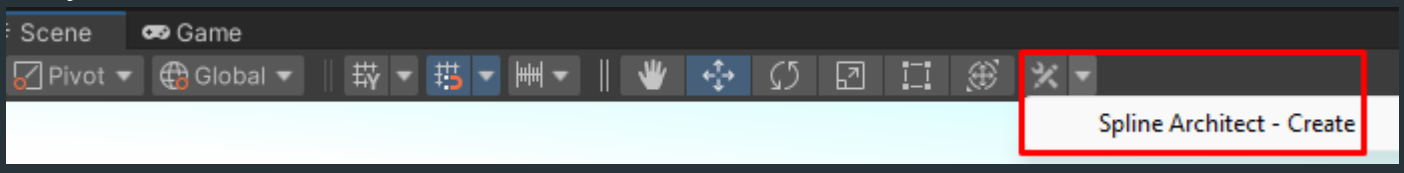
You find the "Spline Architect - Create" button in the regular Tools menu. When its enabled you can either create new splines or new control points on the selected spline.

Delete control points by selecting them and pressing "Delete" or "Cmd + Backspace" on Mac computers.

Unity 6000



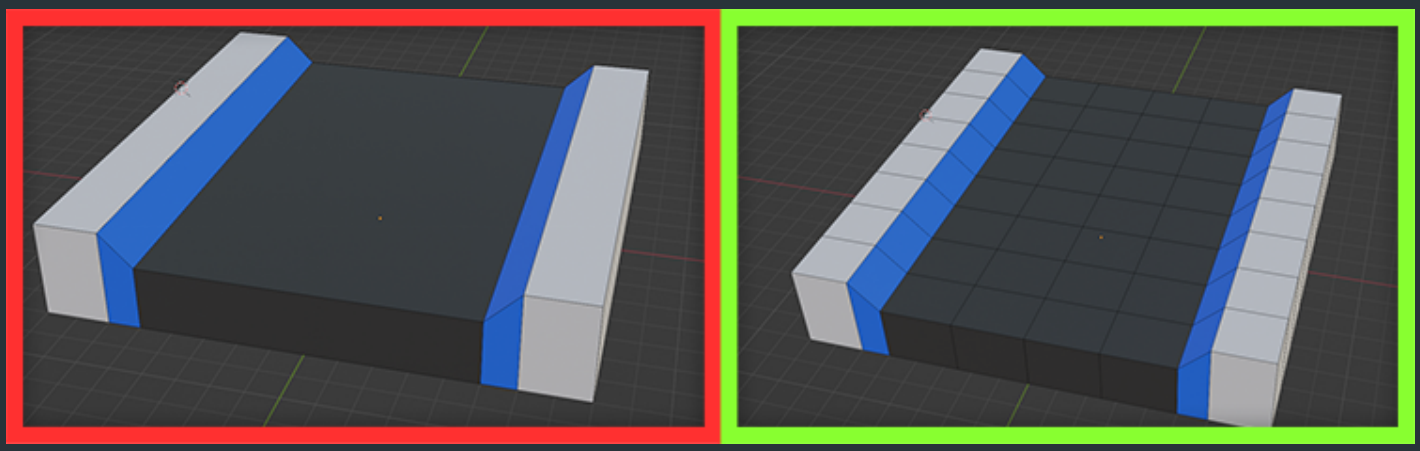
Unity 2022



Create deformation

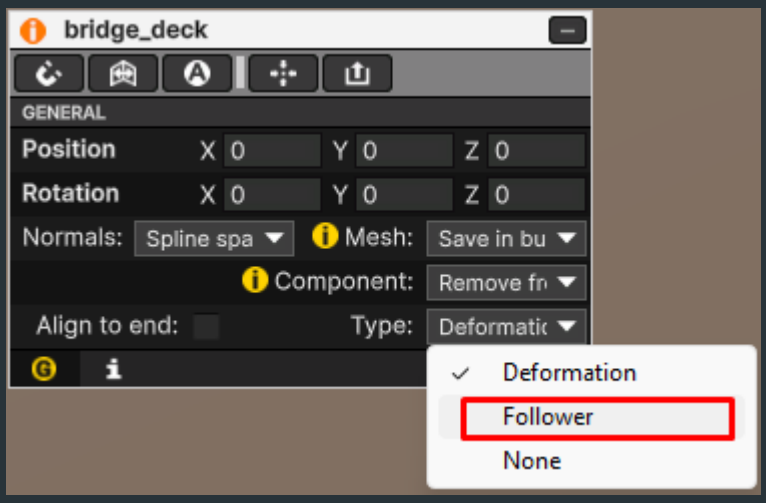
Parent the object to any Spline or Spline Object. This will automatically generate a new mesh derived from the original mesh that the GameObject had in its Mesh Filter or/and Mesh Collider component. Now you can move, delete, duplicate, rotate, and scale the deformation freely on the spline.

Remember to have enough vertices on your meshes. Otherwise, they will not look smooth when deformed.



Create follower

Parent the GameObject to an Spline or Spline Object. After that, select the Game Object and change type to Follower.

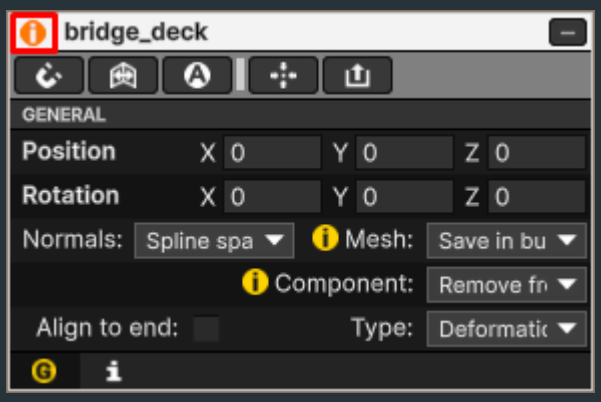


Snapping

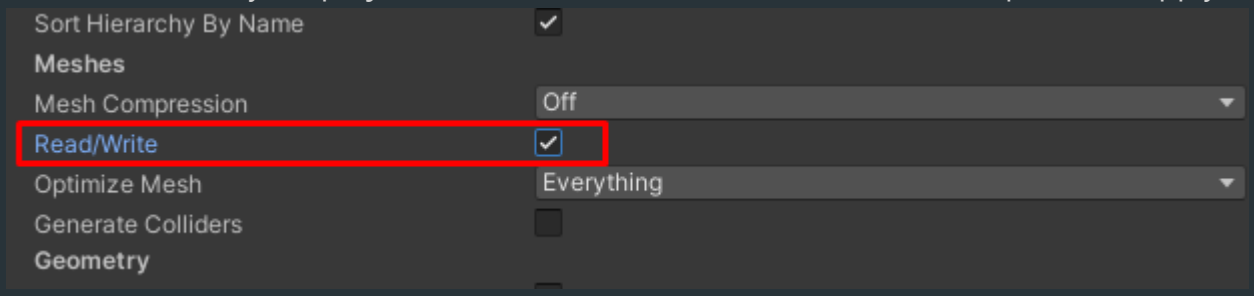
You can snap and align deformations by enable Unity's snapping feature.

Animations

You can animate any deformation or follower using Unity's built in animations system or script your own behaviours. If you see the small orange icon in the upper left corner (in the Spline Object menu), you need to enable read/write access for that mesh (only for deformations). Otherwise, you can't animate it.



Find the mesh in your project folder --> check the Read/Write box --> press the apply button.



> **Note:** Remember that the component option on the spline and SplineObject needs to be set to Active. Otherwise, you won't be able to use them in your built game. More about this below.

Children of Deformations

You can parent a deformation or follower to another deformation, and it will be updated and deformed within that deformation's local space. This can be useful for animating a train with different wagons. You can animate the train using Unity's built-in animation system or through scripting, parent them to the train, and all the wagons will follow.

Note: A deformation can be an empty GameObject.

Children of Followers

Parenting a deformation or follower to another follower will make the object behave like a normal GameObject that ignores the spline space.

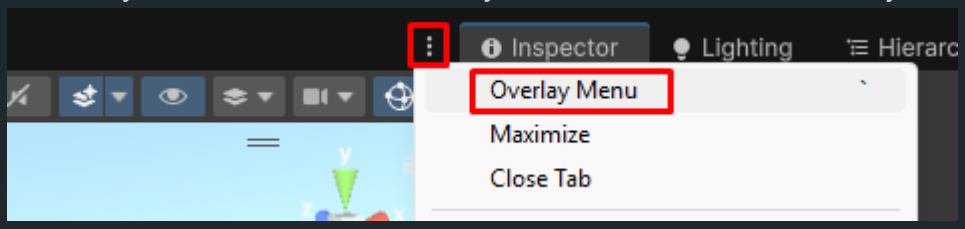
Toolbar/Menus

Spline Architect has its own toolbar, menus and selection system. You have different menus depending on what you have selected. Splines, control points and deformations all have their own with different submenus and settings. Toggle between submenus with the bottom buttons. Some menu items have yellow icons. You can hover over these icons to get information about the specific menu item. You access the menus trough the toolbar, where you also find some other features.

Nearly all functionality in Spline Architect can be found within these menus/toolbar. The exceptions are how you adjust snapping and creating new splines.

Below, we'll go through the toolbar and each menu, and explain all the features.

Note: If you can't see the Toolbar you need to enable it in Unity's overlay menu.



Toolbar

From the toolbar you have access to all Spline Architect menus and some other features.

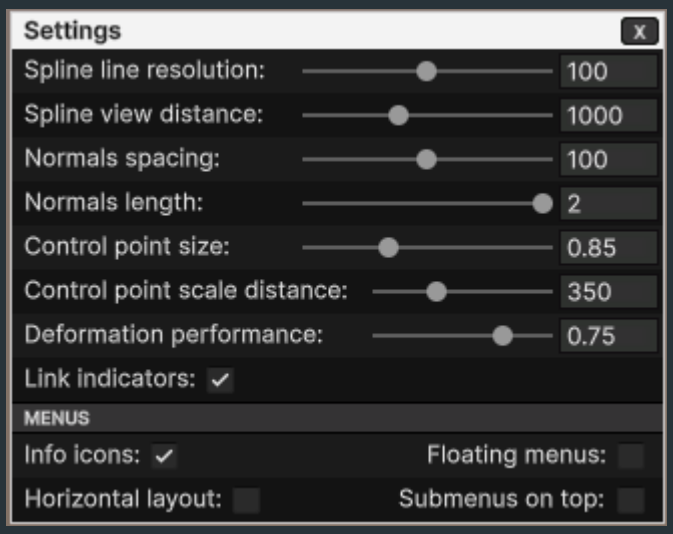
Note: If you are using Unity 2022 you may want to dock the Toolbar because of a visual bug (the bug has no implications on any features).

Buttons:

- **Control panel menus** | Access the spline, spline object and control point menus. You can toggle them by pressing (alt + h).
- **Settings menu** | Access the settings menu that contains various user settings.

- **Info menu** | Access the info menu where you have links to documentations, can see addons installed and more.
- **Grid system** | Toggle the grid system on or off. You also have a small drop down menu where you can change various settings for the grid system.
- **Handle type** | Change between Mirrored, Continuous and broken handle types.
- **Hide splines** | Has 3 modes: show all splines (default), hide all none selected splines, hide all none selected splines and occlude the selected spline. Disabling gizmos will hide all splines, even the selected spline.
- **Show normals** | Toggle normals on or off.

Settings menu



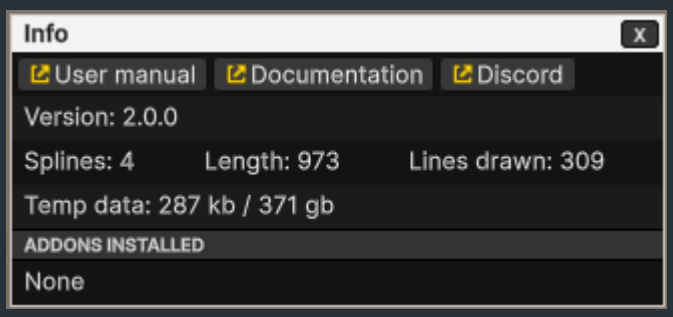
Settings:

- **Spline line resolution** | The resolution on all lines that draws splines.
- **Spline view distance** | Splines will not be drawn beyond this distance.
- **Normal spacing** | How many normals will be drawn for every 100 meters of a spline.
- **Normal length** | The length of the normals displayed along the spline.
- **Control point size** | Change the size of all control points on splines.
- **Control point scale distance** | Sets the maximum distance control points will scale to.
- **Deformation performance** | A higher value will force deformation jobs to run over fewer frames.
- **Link indicators** | Turn link indicators on/off, small dots shown in the Scene view.

Menus:

- **Info icons** | Turn on/off info icons (yellow (i) icons).
- **Floating menus** | Turn floating menus on/off. When floating menus are enabled, you can move all Spline Architect windows freely. This can be useful if you have undocked the Scene view.
- **Horizontal layout** | Determines the layout of the control panel menus.
- **Submenus on top** | Determines whether submenu buttons are displayed at the top or bottom of the window.

Info menu



Info:

- **Version** | Current Spline Architect version.
- **Splines** | How many splines are currently in the open scene(s).
- **Length** | Total length of all splines.
- **LInes drawn** | How many lines Spline Architect will draw to display all splines.
- **Temp data** | Currently only used by the Terrain Tools addon. Displays how much temporary data is being used and how much disk space you have left. Restarting the editor will remove unnecessary data.

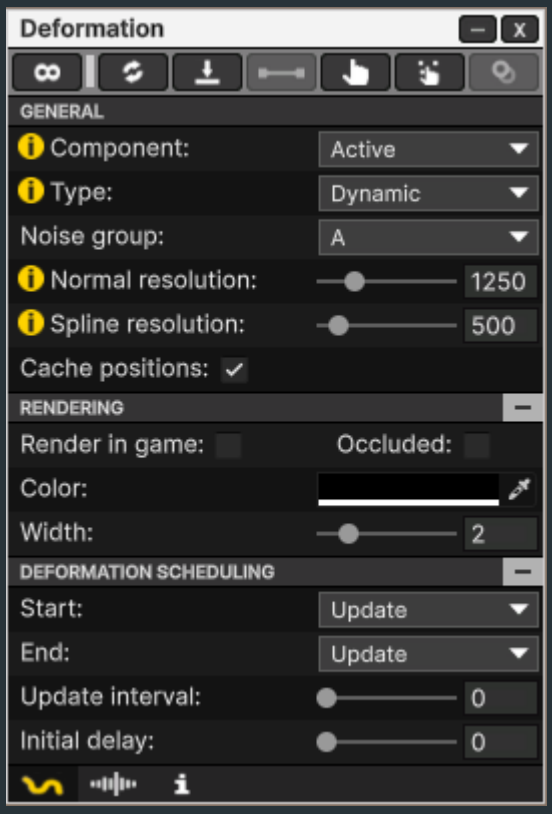
Grid menu

All settings for the grid system.

Buttons:

- **Color** | Change the color of the grid.
- **Size** | Change the size of the grid.
- **Occluded** | Should the grid be occluded or not by other GameObjects.
- **Distance labels** | Show or hide distance labels.

Spline menu - deformation



This panel appears when a Spline is selected, allowing you to configure various settings related to the splines mesh deformation feature.

Top buttons:

- **Toggle loop** | Toggle loop for the selected Spline.
- **Reverse control points** | Reverses the order of all control points.
- **Flatten control points** | Flatten all control points to the same Y value.
- **Align control points** | Aligns all selected control points so it becomes a straight line.
- **Select and center Spline transform** | Selects the Spline's transform and centers it to the center of all control points.
- **Select all anchors** | Selects all the anchors of the Spline.
- **Join selected Splines** | Joins all currently selected Splines.
- **Align grid for selected Splines** | Visible only when "Grid" is enabled. Aligns the grid for all selected Splines so you can work on the same grid across multiple Splines. If only one spline is selected it will center the grid to the center of all control points instead.

General:

- **Components** | How to handle the Spline and SplineObject components during runtime. You can set it to **Active**, **Inactive** or **Remove from build**. If set to **Active** you can use the Spline for various animations or other logic, no spline data will be removed. Setting the spline to **Inactive** will keep the component in your build but removes all spline data during the start function call on the Spline component. Good choose if you want to generate meshes during scene load. Using **Remove from build** will completely remove the component from your built game.

- **Type** | Change between Static 3D, Static 2D, and Dynamic. Static splines don't use cached normals and therefore consume less memory. Their normals remain fixed. Dynamic splines are better suited for complex deformations, such as loops. Static 2D splines are primarily for working in 2D environments, while Static 3D splines are primarily for 3D environments.
- **Noise group** | The noise group you want to be applied to mesh deformations and animations.
- **Normal resolution** | The resolution of cached normals. Only used by dynamic splines.
- **Spline resolution** | The resolution of the spline. If you're working with very long splines you might need to increase this value.
- **Cache positions** | Enabling this option improves overall performance, but the spline will use more memory. In most cases, the memory usage is so small that you don't need to worry about it.

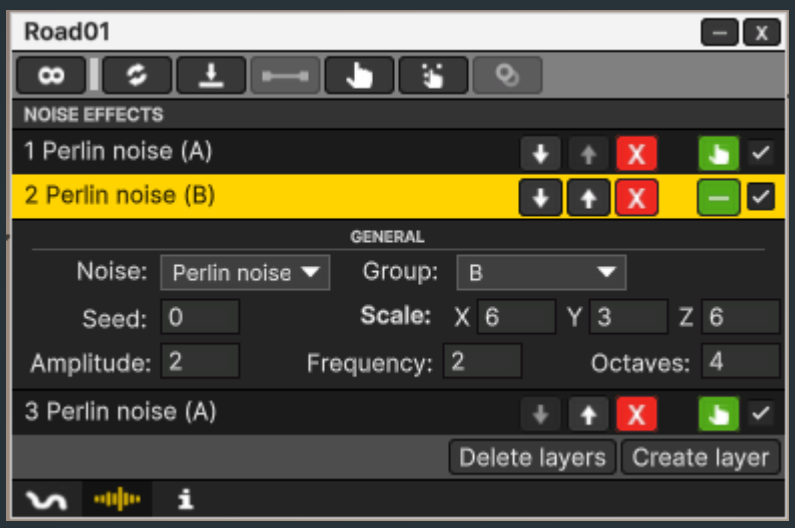
Rendering:

- **Render in game** | Draws the spline in Play mode and in the built game. Useful for debugging when you need to see the spline at runtime.
- **Occluded** | Determines whether the spline is always visible or can be occluded by other GameObjects.
- **Color** | Change the color of the spline's control points and unselected line. The selected spline will always display with a yellow line.
- **Width** | The width of the spline. A larger width also increases the spline's "press area".

Deformation scheduling:

- **Start** | Specifies whether the deformation job starts in Update, Late Update, or is disabled so you can trigger it manually.
- **End** | Specifies whether the deformation job ends in Update, Late Update, or is disabled so you can handle completion manually.
- **Update interval** | The number of frames the deformation job runs before it completes. Increasing this can greatly reduce performance cost and is especially useful for background animations.
- **Initial delay** | The number of frames to wait after the spline is initialized before the deformation job starts. If you want two splines to update on alternating frames, set the Initial Delay to 1 on one of them so they do not complete on the same frame, which can improve performance.

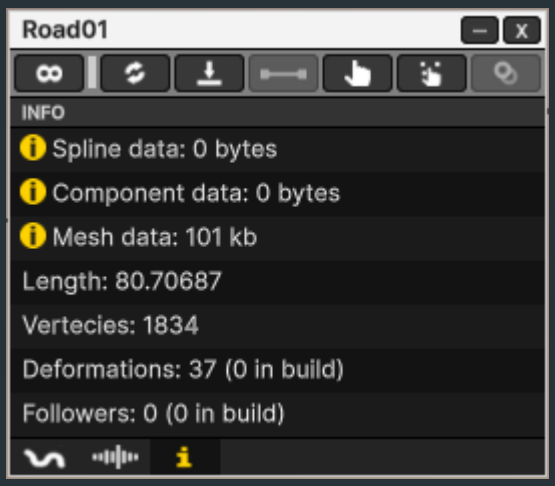
Spline menu - noise



Will appear when a Spline is selected and you have pressed the bottom noise button. Displays all the noise settings that will change the behaviour on mesh deformations and animations.

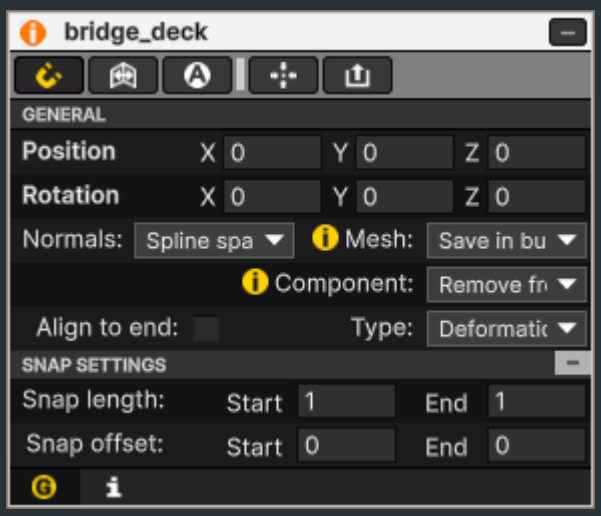
- **Noise** | Choose between 8 different noise effects.
- **Group** | Set the group for the noise layer. You can use the same group for multiple layers and stack noise layers together to create various effects.

Spline menu - info



General info about the Spline, size in memory, length, and so on.

Spline object menu - general



This menu appears when a Spline Object is selected, providing controls for deformation mirroring, centring, and general transformation settings.

Top buttons:

- **Snap** | You can snap your deformations to other objects or the control points on the spline. It has 3 modes: disabled, snap to control points, snap to objects.
- **Mirror deformation** | You can achieve the same deformation effect by scaling negatively, but this approach may cause collision issues in some physics systems. Using the mirror function ensures compatibility across all physics systems without any collision problems.
- **Auto type** | Will automatically switch type between deformation and follower. For example, if you have a road where parts of it is a straight line. All deformations on that straight line will switch to type follower instead and not be deformed. This works really well with the "Align control points" feature and is also very good for performance and memory usage.
- **To center** | Reset the object's X and Y position to 0 and its Euler rotation to (0, 0, 0).
- **Export mesh** | Export meshes for all selected Spline Objects.

General:

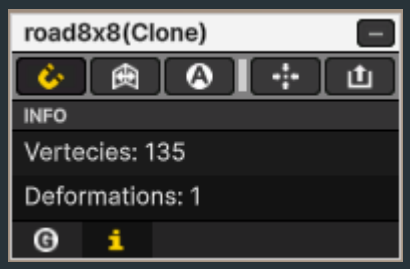
- **Position** | Set the object's position in spline space.
- **Rotation** | Set the object's rotation in spline space.
- **Normals** | Choose between **Spline based (Default)**, **Cylinder based** (good for cylinder shapes), **Unity calculated**, **Unity calculated (Seamless)** (unity default but ignores overlapping vertices) and **Don't calculate**.
- **Mesh** | How to handle the deformed mesh: You can generate it during runtime, save it in the scene file, or save it only in your build. Saving the mesh in your build will reduce the size of your scene files, because the mesh is generated at editor startup (the same applies to all other settings except Save in scene).
- **Lock position** | Only visible when the SplineObject is of type Follower. If this is enabled, the SplineObject will not follow the spline's position but will adjust according to the spline's rotation.

- **Component** | How to handle the SplineObject component during runtime. You can set it to **Active**, **Inactivate after scene load** or **Remove from build**.
- **Align to end** | If enabled, the Spline Object will originate from the end of the spline instead of the start.
- **Type** | Change between Deformation, Follower and None. "None" objects will behave as normal objects and don't interact with the Spline. Followers will just follow the spline and not be deformed. Deformations will be deformed if it has a mesh inside the mesh filter or mesh collider. If set to **Active** you can use the SplineObject for various animations in your built game. Setting the SplineObject's component option to **Inactivate after scene load** will keep the component in your build, but it will be inactivated during the start call. Good choose if you want to generate meshes during scene load. Using **Remove from build** will completely remove the component from your built game. **Note:** You can only change this setting if the Spline Component setting is set to Active. If you have an hierarchy of SplineObjects, you will set the whole hierarchy to the same component value when this value is changed.

Snap settings:
Snap settings can be toggled between three modes: Off, Snap to Control Points, and Snap to a specific point on the spline. You can cycle through these modes by pressing the Snap button multiple times.

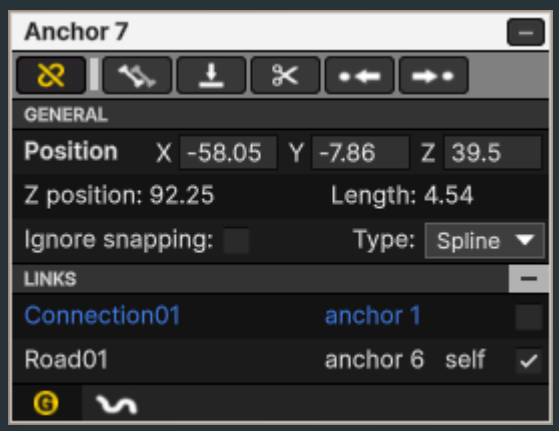
- **Snap length** | How long the deformations start and end points can snap to a control point or object on the spline.
- **Snap Offset** | Offset the snapping position for the start or end of the deformation.
- **Spline point** | Snaps to a specific point on the spline, measured in meters.

Spline object menu - info



General info about the Spline Object.

Control point menu - general



This menu appears when a control point is selected.

Top buttons:

- **Link to closest anchor or spline connector** | Links the selected control point to the closest anchor on any spline or closest Spline Connector.
- **Mirror spline connector** | Only available when a control point is linked to a spline connector. When enabled, the linked control points will use a mirrored direction instead of inheriting the exact direction from the spline connector.
- **Align tangents** | Only available on linked control points, will align all linked control points.
- **Flatten control points** | Flatten only the selected control segment.
- **Split spline** | Split the Spline into two at the selected control point.
- **Next control point** | Move selection to the next control point.
- **Prev control point** | Move selection to the previous control point.

General:

- **Position** | World position for the selected control point.
- **Z position** | The control points position on the spline. The z position is also equivalent to the spline length at that specific point.
- **Length** | The length of the segment.
- **Ignore snapping** | Deformations with snapping enabled with not snap to this control point, if enabled.
- **Type** | Change between Spline and Line type. Creating new control points will have the type of the closest control point.

Links: If you have linked the control point, all links will be displayed in this section. You can also toggle each link on/off, when doing so the link will be ignored when unsing API functions as: spline.FindLinkCrossingsNonAlloc().

Control point menu - deformation

Will appear when a control point is selected and the bottom deformation sub menu button has been pressed. Has different effects that can be applied to deformations.

Other features

Snap/Move control points on object surfaces - Select a control point, hold down Shift + Ctrl, and then click the control point again to activate surface snapping. Now, as you move the control point, it will snap to and follow the surface of any object with a collider.

Create control points using the nearest splines direction - Holding ctrl while creating control points will automatically align the new control point with the closest spline.

Shortcuts

Spline Architect has different shortcuts that can be changed in Unitys shortcut menu. Edit --> Shortcuts...

Components

Spline
Add this component to any GameObject to create an Spline.

Spline Object
This component is automatically added when you attach a GameObject as a child of a spline. You can also add it manually to any GameObject, even if it is not yet placed on a spline, for pre-configuration.

Spline Connector
You can connect a splines control point to the GameObject that has this component. This is useful for animating splines using Unity's animation system or for creating road intersections. Moving the SplineConnector GameObject will update all connected splines.

Mesh pooling

Spline Architect tracks all deformed meshes. For example, if a Mesh Filter and a Mesh Collider on the same Game Object use the same original mesh, only one mesh will be created and used. If you have two different meshes on a collider and mesh filter it will pool two different meshes.